



Major revision requested: 3400076

From Advanced Intelligent Systems <advintellsyst@wiley.com>

Date Tue 9/22/2026 12:03 PM

To Lorenzo Vignoli <lorenzo.vignoli@epfl.ch>

WILEY



Dear Mr. Vignoli,

Thank you for submitting your manuscript "Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms" (ID 3400076), to Advanced Intelligent Systems.

After careful assessment and based on the reviewer reports, we have decided your manuscript requires major revisions in order to proceed with the peer review process and be considered for publication. Reviewer comments are included at the bottom of this email to assist you in revising your manuscript.

Please take the following points into account during your revision:

Please ensure that the title of your manuscript is succinct, descriptive, and accurate, so that readers may easily find and cite your work by simply searching for relevant keywords

Reviewers may have suggested references to be added to the manuscript. While this may be a reasonable request to improve your manuscript, please note that there is no obligation to include these specific references. Please include a response to such requests as part of your revision letter.

Manuscripts generally benefit from careful re-reading before submitting the revised version. We also recommend running a spell check. Please define all acronyms when first mentioned and apply them thereafter (the title, abstract, and Table of Contents text must each be understandable as standalone texts). Check that the text and axis labels in all graphics are correct and are correctly spelled. In addition, please check that all parts, insets, and footnotes of Figures and Schemes are mentioned in the caption.

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We ask that you return your revised manuscript by October 13, 2026. If you need more time, please let us know. We are happy to discuss an extended deadline.

When submitting your revised manuscript, you will be asked to upload an author response letter that details the revisions you have made in response to editor and reviewer comments. You may also be required to show revisions made within the manuscript files. Please follow any journal-specific guidelines for formatting and submitting revisions outlined in the author guidelines page.

When you have finished revising your manuscript, follow the link below to submit your revision:

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Thank you for submitting your work to Advanced Intelligent Systems. We look forward to receiving your revised manuscript.

Kind regards,
Dr. Richard Murray
Advanced Intelligent Systems

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The following reviewer comments were taken into consideration during the peer review process.

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Reviewer 1 Comments to the Author

Overall assessment

This manuscript presents an internal camera-based framework for pose estimation, model-based backbone reconstruction, and quasi-static control of tendon-driven soft robotic arms. Integrating miniature cameras into one- and three-section Trimmed Helicoid manipulators is technically interesting, and the comparison between model-based

and learned inverse-kinematics controllers is potentially valuable.

I recommend major revision. The claimed contributions (internal-camera proprioception, learned inverse kinematics, closed-loop soft-arm control, and redundancy resolution) have each been extensively addressed in prior work. The originality of the paper therefore

appears to lie primarily in their particular integration: distributed markerless internal cameras combined with model-based and learned redundancy-aware control on a multisection arm. The authors should define this contribution more precisely and provide direct quantitative comparisons or demonstrate a capability not established previously.

Several central claims also extend beyond the evidence presented. The full-body curvature estimate is not used for control; scalability is demonstrated only with separately trained one- and three-section networks; multi-section curvature labels rely on a learned surrogate rather than direct full-body measurements; and the manuscript reports no quantitative experimental results isolating the effect of the secondary objective. In addition, the control evaluation uses a limited set of targets and planar trajectories without repeated trials. The related-work discussion, reproducibility details, curvature formulation, and figures require significant improvement.

Major comments

1. Clarify the role of visual sensing in closed-loop control

The manuscript presents pose estimation and full-body piecewise-polynomial-curvature (PPC) reconstruction as parts of a unified sensing-and-control framework. However, Algorithms 1 and 2 and Fig. S3 indicate that both controllers use feedback from the pose network, while the curvature network is evaluated separately and does not enter either control law.

Algorithm 1 also uses a PCC inverse calculation to initialize tendon lengths from section-tip positions. This is not the same as using the image-based PPC curvature estimate. Moreover, the resulting estimate appears to be replaced by the motor-encoder measurement at the first control iteration, before it affects the control command.

The authors should explain this initialization or remove it.

The abstract, contribution list, Fig. 4c, and conclusion should clearly distinguish:

camera-based pose estimation used for control;

encoder measurements used to calculate the Jacobian and null-space term; and

image-based PPC reconstruction evaluated as a separate sensing capability.

If the curvature estimate is intended to contribute to control, the manuscript should explicitly show where it enters the control law.

2. Establish originality claims through direct comparison with prior work

The individual sensing and control elements of this paper have appeared in many papers. Novelty can't be based solely on combining internal vision, learned kinematics, and closed-loop control at a general level. The paper's contribution appears to lie more specifically in integrating distributed markerless internal cameras with model-based and learned redundancy-aware control on a multisection arm. This contribution should be defined precisely and compared directly with prior systems.

Several particularly relevant studies are missing or insufficiently discussed:

Hofer et al. (2021), A Vision-Based Sensing Approach for a Spherical Soft Robotic Arm, integrated cameras into three soft actuators, used a CNN to estimate arm orientation, and demonstrated camera-based closed-loop control. The paper reported approximately 1 deg orientation accuracy at 30 Hz, compared with the present manuscript's 3.11 ± 1.96 deg single-section error and 8 Hz update rate.

Thuruthel et al. (2017), Learning Closed Loop Kinematic Controllers for Continuum Manipulators in Unstructured Environments, demonstrated learned inverse kinematics with feedback on a redundant tendon-driven continuum arm, including evaluation under external loads.

Han et al. (2025), Anchoring Morphological Representations Unlocks Latent Proprioception in Soft Robots, used an internal camera to estimate full-body deformation and force.

Lu et al. (2024), Adaptive Online Learning and Robust 3-D Shape Servoing of Continuum and Soft Robots in Unstructured Environments, is context for learned shape servoing and robustness, although it uses FBG sensing.

Yoo et al., Toward Zero-Shot Sim-to-Real Transfer Learning for Pneumatic Soft Robot 3D Proprioceptive Sensing, has internal-camera soft-robot proprioception.

The authors should explain what each close prior study established and identify the specific capability added here. The claim that no prior framework combines comparable sensing with closed-loop null-space control is too broad without this analysis.

3. Substantiate or narrow the scalability claim

The estimator is not invariant to the number of arm sections. Images from n cameras are stacked as n input channels, so the first convolutional layer changes with n . The output head must also change—from 6 to 18 pose variables and from one to three sets of curvature parameters. Separate networks are trained for the one- and three-section arms.

Transfer learning may reduce retraining effort, but the present method is not plug-and-play across arbitrary arm lengths and is not validated on an unseen section count. The statement that only the first layer must be adjusted also conflicts with Fig. 3 because the regression head changes as well.

4. Strengthen the ground truth for full-body reconstruction

For the single-section arm, curvature labels are fitted from three measured orientations. For the multi-section arm, SI Section 4.2 states that marker occlusion prevents simultaneous full-body tracking and that a learned network, N_C , maps tip pose to curvature labels. Thus, the multi-section CNN is evaluated primarily against a learned, model-constrained surrogate rather than direct measurements of the backbone.

Fig. S2b does not provide sufficient validation of N_C : numerical errors, sample count, data split, and confidence intervals are missing. It is also unclear whether its validation configurations are independent of the data used to train N_C and the three-section curvature estimator.

Wherever possible, reconstructed backbone points should be compared directly with motion-capture marker positions, and uncertainty in N_C should be propagated into the reported errors. Otherwise, the result should be described as model-constrained reconstruction, with its assumptions stated explicitly, including planar bending within each section, no torsion, and no arc-dependent bending direction.

5. Correct the curvature-coordinate inconsistency

The main manuscript defines s as normalized arc length and writes $\theta(s) = c_0 s + (1/2)c_1 s^2 + (1/3)c_2 s^3$ in Eq. 4. Under this definition, initialization should not introduce factors of section length L . In contrast, SI Eqs. S5–S9 evaluate the polynomial at $L/3$, $2L/3$, and L , and use $c_0 = \theta/L$ for constant curvature. This treats the independent variable as dimensional arc length.

These formulations have different coefficient units and cannot both be correct. The paper should consistently use either normalized arc length $u = x/L$ or dimensional arc length $x \in [0, L]$, and revise Eqs. 4–6, SI Eqs. S3–S9, the translation integral, nomenclature, and coefficient units accordingly.

6. Quantitatively evaluate the secondary objective

The control contribution is framed as redundancy-aware null-space control, but the manuscript reports no quantitative experimental results isolating the effect of the secondary objective. No reported experiment shows the objective $H(l)$, tendon-limit margins, tendon distributions, or configurations obtained with and without the secondary objective. Furthermore, $H(l)$ penalizes displacement from the mean sampled tendon vector. It does not directly minimize tendon force, motor current, energy, or control effort. The objective should therefore be described as actuator-range centering unless effort is measured.

For both controllers, an ablation with the secondary objective disabled using identical targets would be helpful. Results should report both primary-task error and a quantitative secondary-task

metric. A useful experiment could compare multiple configurations reaching the same tip pose and show the resulting tendon margins or values of $H(l)$.

7. Expand and better characterize the control evaluation

The experimental evaluation does not quantify its coverage relative to the reachable workspace. The training datasets reportedly include bending angles up to 85° per section, and the multisection target plots span a substantial positional range. However The trajectory evaluation is narrow, and the target coverage is incompletely documented. Single-section bending angles are apparently under 17° , and three-section angles are under 19° , and overall length changes are also quite limited. The evaluation does not include workspace-boundary paths, nonplanar trajectories, large orientation changes, or trajectories requiring qualitatively different section configurations.

Figs. S4–S6 plot the target positions and associated errors, but the coordinates are not tabulated and the targets are not compared with the complete reachable or sampled workspace. The authors should report the target coordinates, associated bending and compression ranges, and the fraction of the sampled workspace represented by the evaluation set.

Each of the 20 targets is visited only once. Consequently, the reported standard deviations describe variation across different targets, not trial-to-trial repeatability, and do not establish that one controller outperforms the other. The 30 waypoints in each trajectory are likewise serially correlated rather than independent trials.

The disturbance experiment should also be characterized more narrowly. A single 50 g off-center payload demonstrates compensation for one constant load (which is only moderately successful), not general robustness to disturbances or contact. Either additional load magnitudes and directions should be tested or the claims should be limited.

8. Establish independence of training and evaluation data

The datasets are randomly divided 85/10/5, but the acquisition sequence, session structure, and grouping of neighboring configurations are not described. If frames from continuous trials are split randomly, nearly identical or temporally adjacent configurations may appear in both training and test sets, inflating apparent generalization.

The authors should state:

how samples were randomized;

whether test data were separated by session or configuration neighborhood;

whether controller targets and trajectories were excluded from training; and

whether experiments span different days, camera settings, coverings, or robot builds.

9. Provide the missing reproducibility details

The paper needs to include enough information to enable reproduction of the results. The methods omit several parameters that affect performance: camera model, lens and field of view, exposure and gain behavior, maximum epochs, random seeds, output normalization statistics, angular decoding, and the training details for N_C , N_{p3} , and N_{I3} . Additionally, robot design details are also incomplete, including tendon material and diameter, preload/zeroing criterion, current limit, sampled tendon ranges, and the number of acquisition sessions. The SI is a good place for a single reproducibility table containing these settings. Claims of robustness to lighting or hardware variation should be limited unless such variation was explicitly tested.

10. Revise the figures and captions

Several figures—particularly Figs. 1B, 2, 3, and 5—are difficult to interpret. Their visual density and inconsistent presentation make it hard to follow the methodology and compare results. In particular, Figure 3 requires attention. Although the MobileNetV2 backbone and inverted residual block are depicted, the overall relationship among the pose, curvature, forward-kinematics, and inverse-kinematics networks is hard to reconstruct. The MLP inputs and outputs are not identified, and tensor dimensions and network-specific outputs are unclear. The paper should use a consistent flow direction, label all network inputs and outputs, and distinguish clearly between the networks used for pose estimation, curvature reconstruction, forward kinematics, and inverse kinematics.

Minor Comments

Pages 1-2

Use 'markerless at measured inferences' rather than simply 'markerless.' External optical markers are required to generate the supervised labels, even though markers are not required during deployment.

Use either 'section' or 'segment' consistently throughout the manuscript, figures, algorithms, and SI.

Pages 3-5

The calibration paragraph says that 'all ten tendons' are tightened, but the multi-section arm has nine tendons and the single-section arm has three. Correct the number and state whether the same preload procedure is used for both systems.

Identify the camera manufacturer and model, lens or field of view, nominal frame rate, and the physical size and mass of each camera and mount.

The claim that images are acquired simultaneously should be supported by the synchronization method as this type software capture

is not necessarily simultaneous hardware triggering.

Pages 5-7

Explain how angular sine/cosine outputs are converted back to angles and how losses for position, orientation, length, and curvature are balanced after normalization.

Clarify the negative sign in Eq. S11 and clarify how the highlighted pixels are the regions used for prediction.

In Algorithm 1, the initial line that estimates tendon lengths from intermediate positions via PCC appears unused because tendon lengths are read from encoders inside the loop. Remove it or explain its role.

Use consistent Jacobian notation. J_{t2l} , J_{l2p} , and their learned representations currently alternate between a forward Jacobian, an inverse Jacobian, and a derivative of an inverse network. These might be confusing to follow for the readers.

State whether the open-loop baselines use the same measured initial pose, planner, gains, stopping limits, and commanded waypoint sequence as the closed-loop methods. 'Open-loop' and 'feedforward' should be used consistently.

Pages 8-11

State the denominator used for every percentage error. For example, clarify whether multi-section position errors are normalized by one section length, cumulative tip length, or the maximum deployed length.

Figure 5b intentionally uses nonuniform axis scaling. This visually distorts the reconstructed shapes. Use equal spatial scaling where possible or add a second undistorted view and explicit scale bars. Figure 7 alternates among 'cumulative 3D error,' 'sum of 3D errors,' and 'average 3D error for each point.' Define the statistic exactly. Summing three points for the single-section arm and nine for the multi-section arm also prevents direct comparison. Per-point RMSE or error versus normalized arc length would be clearer.

In Table 1, the text states that orientation error remains below 9 degrees, but several standard deviations are larger than the corresponding means. Make clear that only the mean is below 9 degrees and report median, 95th percentile, and maximum error. The claim that bent configurations improve accuracy is plausible but currently qualitative. The authors should plot error against bending angle before drawing this conclusion.

For Figs. 6, 9, and 10, identify the start direction, waypoint order, number of completed cycles, and whether the plotted paths are one run or an average. Add quantitative path errors in the caption or an accompanying table.

The discussion should distinguish demonstrated capabilities from proposed applications. Obstacle avoidance, whole-body grasping, surgical guidance, and force or contact sensing are future directions and were not evaluated here.

Supporting Information

Fig. S2b should report the numerical N_C error for each curvature model, the number of validation configurations, and whether the bars show standard deviation, standard error, or another interval.

Table S6 contains highly skewed errors, particularly 11.92 +/- 25.10 mm for Section II and orientation standard deviations near 16 degrees. Report robust distribution summaries and discuss the outliers rather than relying only on mean and standard deviation.

Figs. S4-S6 should use matched target identifiers or connecting pairs so the reader can compare the same workspace locations across controllers. Use consistent color limits when the goal is direct visual comparison.

Tables S3 and S4 list dataset fields but not dataset composition. Add sample counts per acquisition mode and robot configuration, including how many samples were removed and why.